

Peak Shaving



Introduction

Electricity customers usually have an uneven load profile during the day, resulting in load peaks. The power system has to be dimensioned for that peak load while during other parts of the day it is under-utilized. The extra costs in keeping up with the peak demand are passed to the customers in form of a power fee, i.e. you pay for your maximum peak load. By utilizing Peak shaving, peak load can be reduced and hence the power fee. System is controlled to charge up during off-peak hours and discharged during peak hours. Households' peak loads often coincide with the peak load of the overall grid. That means the cost of energy is also high during these times. In such cases the benefit of peak shaving is double by reducing both the power fee and the cost of energy.

Advantages

Cost-saving

Maximize PV energy to save more on electricity bill

Eco-friendly

Clean solar energy utilization

Flexible

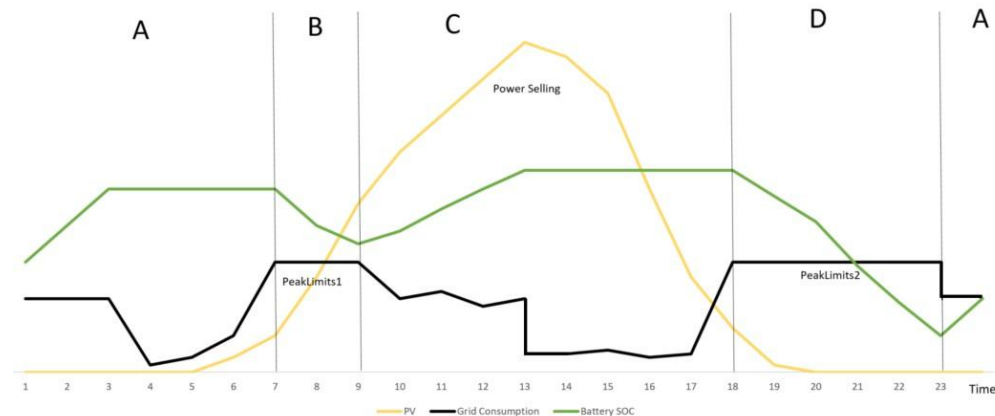
Set period flexibly to meet complex scenarios

Compatibility

Function	Inverter Type
Peak Shaving	X1-IES
	X3-IES
	X1-Hybrid-G4
	X3-Hybrid-G4
	X1-Fit-G4
	X3-Fit-G4
	X1-AC
	X1-Vast 5-10kW
	X3-HYB-G4 PRO
	X3-ULTRA
	A1-AC-G2
	J3-ULTRA-LV
	J3-ULTRA

Working Modes

With Peak shaving function, SolaX inverter will be your smart home energy manager to control the overall usage of energy in the house and maximize energy self-consumption in a smarter way.



There are overall four operation period that you can choose from:

Period A: Charge from Grid Period

This period allow inverter to take energy from grid to charge battery in order to have enough backup for peak shaving. Please note that this period start from ShavingEndTime2, end until ShavingStartTime1.

- Enable/Disable

Allow inverter to take grid energy to charge battery or not.

- ChargePowerLimits

Settable target power from grid. Inverter will use this target power to charge battery from grid. (1kW by default)

- MaxSOC

Inverter will take grid energy to charge battery until SOC reach this value. (50% by default. 10%~100% settable)

Period B: Peak Shaving Period1

Shaving period should be set to cover load peaks. Battery will be discharged to shave load peak until battery SOC drop to Min SOC(10% by default).

- PeakLimits1

Once the consumption (grid side) reach this value, inverter will start shaving to keep the consumption below this value.

- ShavingStartTime1

Inverter allow battery discharging to shave consumption from this time.

- ShavingEndTime1

Inverter allow battery discharging to shave consumption until this time.



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Period C: PV Charge Period

In this period, inverter not allow taking grid energy to charge battery. PV is the only way to recharge battery and **PV will charge the battery first, not the load first.**

Reserved SOC

Inverter will not supply power to loads until battery SOC higher than this value in order to save enough energy for later shaving period. (50% by default. 10~100% settable)

Period D: Peak Shaving Period2

Same with above Period B: Peak Shaving Period1.

- PeakLimits2
- ShavingStartTime2
- ShavingEndTime2

Typical Case

- ① Consumption exceeds 5kW will be charged with high electricity price during 7:00~9:00.
- ② Consumption exceeds 7kW will be charged with high electricity price during 18:00~23:00.
- ③ Electricity price fluctuates sharply at different times every day:
Expensive at peak time 7:00~9:00 and 18:00~23:00. Cheap at valley time 00:00~3:00.

Settings example(comply with above case):

- PeakLimits1 : 5kW
- ShavingStartTime1 : 7:00
- ShavingEndTime1 : 9:00
- PeakLimits2 : 7kW
- ShavingStartTime2 : 18:00
- ShavingEndTime2 : 23:00
- ChargeFromGrid : Enable
- ChargePowerLimits : 3kW
- MaxSOC : 50% (Take energy from grid only to charge battery to 50% during ShavingEndTime2~ShavingStartTime1 23:00~7:00. 50% limitation only limit the energy from grid, not limit PV)

- Reserved SOC : 50% (This settings only impact during period C
ShavingEndTime1~ShavingStartTime2 9:00~18:00, PV will not supply loads until SOC
higher than 50%)